

ANALYSIS

Open Access



DENND2D serves as a novel prognostic biomarker with paradoxical protumorigenic effects in glioma

Hongxing Ye¹, Chao Zhang¹, Linghao Bu¹, Feng Xiao¹, Yuxiang Weng¹, Kaiyuan Huang¹, Zebin Fang¹, Weijian Fan¹, Hao Jiang¹, Yu Zhu¹, Renya Zhan¹, Xiujue Zheng¹, Yumin Wang^{2,3}, Zhouying Peng^{2,3*} and Luyuan Zhang^{1*}

*Correspondence:

Zhouying Peng
zypeng@csu.edu.cn

Luyuan Zhang
1320110@zju.edu.cn

¹Department of Neurosurgery, The First Affiliated Hospital, Zhejiang University School of Medicine, Hangzhou, Zhejiang, China

²Department of Otolaryngology Head and Neck Surgery, Xiangya Hospital, Central South University, Changsha, China

³National Clinical Research Center for Geriatric Disorders, Xiangya Hospital, Changsha, China

Abstract

Introduction Glioma, especially glioblastoma (GBM), has a dismal prognosis with current therapies. In this study, we aimed to identify potential novel molecular targets for GBM by analyzing RNA sequencing.

Methods RNA-seq data from TCGA, CGGA, and GTEx were analyzed to assess *DENND2D* expression across glioma grades and molecular subtypes. Survival analysis, methylation/copy number profiling, weighted gene co-expression network analysis (WGCNA), functional enrichment (GO/KEGG), and immune infiltration deconvolution (CIBERSORT) were performed.

Results By multi-cohort analysis, 5 upregulated genes and 23 downregulated genes demonstrated prognostic significance in GBM. *DENND2D* was upregulated in high-grade gliomas (GBM > LGG; $p < 0.05$) and correlated with poor prognostic features (IDH-wildtype, non-1p/19q-codeletion). High *DENND2D* levels predicted worse survival (HR > 1, $p < 0.001$) in TCGA and CGGA cohorts. Expression was inversely correlated with promoter methylation (sites cg00619207/cg24641737) and was modulated by copy number alterations. WGCNA revealed co-expression with immune regulators (RAC2/TLR2/ITGB2/PTPRC/HLA-DPA1) and enrichment in leukocyte activation pathways ($p < 0.01$). CIBERSORT showed that *DENND2D*^{high} tumors had elevated immunosuppressive infiltrates: M1/M2 macrophages and $\gamma\delta$ T cells in LGG, and M2 macrophages in GBM ($p < 0.05$).

Conclusion *DENND2D* acts as an oncogene in glioma, driving aggressiveness and shaping an immunosuppressive microenvironment. It represents a novel prognostic biomarker and therapeutic target for overcoming immunotherapy resistance.

Keywords *DENND2D*, Glioma, Prognostic biomarker, Tumor microenvironment, Immunosuppression



1 Introduction

Gliomas are the most common primary tumors arising from neuroglial cells. Grade IV glioblastoma (GBM) is the most aggressive type, conferring a devastating prognosis with a median survival of less than 15 months [1–3]. High invasiveness, tumor heterogeneity, and the initiation of cell dormancy drive rapid progression, frequent recurrence, and poor outcomes. Currently, the survival rate of patients with GBM is much lower than that of other patients with low-grade glioma, which has become an important problem that needs to be urgently addressed.

Standard care involving surgery, radiotherapy, and chemotherapy remains nearly universal, yet glioma remains largely incurable. Targeted therapies and immunotherapies represent major research areas [4, 5]. Despite promising potential, their efficacy remains limited, highlighting the critical need for novel therapeutic targets. In recent years, the development of sequencing technology has revolutionized the molecular pathological diagnosis of glioma. By comprehensively analyzing its genomic, transcriptomic, and epigenetic characteristics, a key basis can be provided for personalized targeted treatment strategies for glioma [6]. However, research reports on the combined analysis of multiple databases for GBM are still lacking and not mature enough.

To identify potential novel targets for GBM, we analyzed RNA-seq data from The Cancer Genome Atlas (TCGA) and the Chinese Glioma Genome Atlas (CGGA) datasets to investigate the aberrant expression of *DENND2D* (DENN Domain Containing 2D) in glioma and its correlations with clinicopathological features, molecular subtypes, patient prognosis, and infiltrating immune cell subpopulations. Intriguingly, our findings revealed that, in contrast to previous studies in other cancers, *DENND2D* functions as an oncogenic factor in glioma, is associated with poorer prognosis, and may exert its effects through immune modulation. These results elucidate the role and functional mechanisms of *DENND2D* in glioma and offer novel insights for the development of future glioma therapies.

2 Materials and methods

2.1 Data collection

Transcriptomic, SNP, and methylation data for glioma (GBM and LGG) cases, along with corresponding clinical metadata, were downloaded from TCGA. Due to the lack of normal brain tissue or adjacent non-tumorous brain tissue genomic data in TCGA, normal brain tissue transcriptomic data were acquired from the Genotype-Tissue Expression (GTEx) project. Transcriptomic data for glioma cell lines were obtained from the Cancer Cell Line Encyclopedia (CCLE). Additional genomic and clinical data of patients with glioma were retrieved from the Chinese Glioma Genome Atlas (CGGA). Supplementary TCGA genomic and clinical metadata were obtained through the cBioPortal platform [7].

All transcriptomic data were normalized to Transcripts Per Million (TPM). This study incorporated 1,671 normal brain tissue samples from GTEx (selected from an initial cohort of 2,076 samples by excluding 405 cases with missing values), 56 glioma cell lines from CCLE, 154 TCGA-GBM samples, and 511 TCGA-LGG samples. All datasets maintained TPM normalization consistency throughout the study. Differentially expressed genes (DEGs) were identified using a threshold of $|\log_2(\text{fold change})| > 1$ (equivalent to fold change > 2 or < 0.5) with adjusted p-value < 0.05 to account for multiple testing.

HTSeq-Counts downloaded from TCGA data were normalized to TPM. The transformation steps were as follows: (1) the read counts were divided by the length of each gene in kilobases to obtain reads per kilobase (RPK), (2) all RPK values in a sample were summed and divided by 1,000,000 (per million scaling factor), and (3) the RPK values were divided by this per million scaling factor. In copy number variation (CNV) data, a $\text{Segment_value} > 0.2$ was defined as amplification, and a $\text{Segment_value} < -0.2$ was defined as deletion. The R package *mgmtstp27* was used to predict the promoter methylation status of TCGA-GBM and TCGA-LGG. Glioma survival time was defined as the time between initial diagnosis and death or last follow-up. Because TCGA CNV probes did not cover the whole chromosome, 1p/19q codeletion status was derived using copy number data, as previously described. All the above operations were implemented in Python.

2.2 Identification of DEGs

A two-sample t-test was used to analyze DEGs. Differences were considered significant if the p-value was < 0.05 . Concurrently, genes exhibiting a fold change (FC) > 2 or < 0.5 were considered differentially expressed. Transcriptomic data from central nervous system (CNS) tissues obtained from GTEx served as the normal brain tissue reference. DEGs were identified by comparing these reference data against: (1) glioma cell line transcriptomic data from the CCLE, (2) the TCGA-LGG dataset, and (3) the TCGA-GBM dataset. Genes demonstrating significant differential expression (adjusted p-value < 0.05) in all three comparisons were designated as core DEGs. Subsequently, we independently performed DEG analysis on the TCGA-LGG and TCGA-GBM transcriptomic datasets (each compared to the GTEx normal CNS reference). Overlapping DEGs common to both the TCGA-LGG and TCGA-GBM analyses were identified and defined as hub genes for subsequent investigation.

2.3 Survival analysis

The median value of gene expression was chosen as the cutoff to divide patients into high- and low-expression subgroups. A log-rank test (p-value < 0.05) was considered significant. A Kaplan-Meier (K-M) plot was used to visualize the association between gene expression and overall survival (OS) time. A Cox hazard regression model with a p-value < 0.05 was also considered significant. A hazard ratio (HR) > 1 was considered a risk factor and an HR < 1 was considered a protective factor. Genes that were significant in both Cox regression and log-rank tests, and that also appeared in the overlapping gene set, were selected as hub genes.

2.4 Methylation and copy number analysis

TCGA glioma patients were divided into high- and low-expression subgroups according to the hub gene expression levels. Student's t-test was used to evaluate methylation data in the two subgroups. Because of the lack of normal samples in TCGA, we used the Human Disease Methylation Database (DiseaseMeth 2.0) [8] to analyze hub gene methylation levels between tumor samples and normal tissue. Moreover, MEXPRESS was used to visualize the association between hub gene methylation, gene expression, and clinical data.

2.5 Gene enrichment analysis

Weighted correlation network analysis (WGCNA) was implemented to analyze the co-expression network. The expression values were transformed using $\log_2(n+1)$, where n is the expression value of each gene. The Pearson product-moment correlation coefficient between each gene was calculated. R-square was set to 0.85. The PARPi-associated genes were clustered into several modules, and the correlation between the modules were also calculated. The topological overlap matrix (TOM) was calculated, and a TOM plot was constructed for visualization. According to the correlation coefficient between the module and the phenotypic data, modules with a correlation coefficient ≥ 0.2 were selected to analyze the correlation of in-module genes with phenotypic data. The module with the highest correlation coefficient between gene significance and module membership was identified as the key module. All of the above operations were performed using WGCNA in R.

Gene Ontology (GO) enrichment and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway analyses were performed using the R package clusterProfiler. GO terms were divided into cellular components, biological processes, and molecular functions. GO and KEGG terms with adjusted p -value < 0.05 were considered significant. These terms were visualized using the GOplot package. To investigate the correlation between *DENND2D* expression levels and glioma immune cells, we estimated immune cell proportions using CIBERSORT. A total of 22 immune cell types were included in this study. The correlation between *DENND2D* expression and these immune cells was also analyzed. TCGA-LGG, TCGA-GBM, and CGGA were analyzed separately. We analyzed the correlations between *DENND2D* expression and tumor purity, ESTIMATE score, immune score, and stromal score. To ensure the reliability of the results, a comprehensive analysis was conducted using the TIMER, xCell, and MCP-counter algorithms. This multi-algorithm approach was employed to validate the findings from CIBERSORT.

2.6 Tissue immunohistochemistry and clinical imaging

Tumor specimens from patients with GBM were obtained from the First Affiliated Hospital, Zhejiang University School of Medicine. IHC and HE staining were performed as previously described [9]. All sections were stained with primary and secondary antibodies, and pathologists independently selected representative fields. The primary antibody used for the immunoassays was against *DENND2D* (Thermofisher, Catlog# ab184799). Preoperative MRI images corresponding to the pathological samples were also obtained. The patient's pathological specimen was diagnosed as a glioma by the First Affiliated Hospital, Zhejiang University School of Medicine.

3 Results

3.1 Differential gene analysis

Comparative analysis revealed 13,174 DEGs in glioma cell lines (CCLE vs. GTEx: 7,464 upregulated and 5,710 downregulated), 15,194 DEGs in primary glioblastoma (TCGA-GBM vs. GTEx: 11,742 upregulated and 3,452 downregulated), and 11,673 DEGs in lower-grade gliomas (TCGA-LGG vs. GTEx: 8,370 upregulated and 3,303 downregulated). A core signature of 2,411 genes demonstrating universal upregulation across all tumor-normal comparisons was designated TN-UP (Tumor vs. Normal Upregulated),

while 1,507 consistently downregulated genes constituted the TN-DOWN (Tumor vs. Normal Downregulated) signature (Fig. 1a).

Analysis of TCGA-GBM vs. TCGA-LGG identified 6,946 DEGs, with 5,253 upregulated (476 overlapping with TN-UP signature) and 1,693 downregulated (344 overlapping with TN-DOWN signature) genes. Similarly, a comparison of CGGA-GBM vs. CGGA-LGG revealed 2,026 DEGs, comprising 1,244 upregulated (326 overlapping with TN-UP) and 782 downregulated (196 overlapping with TN-DOWN) genes. Cross-comparison filtering yielded 302 consistently upregulated genes and 161 consistently

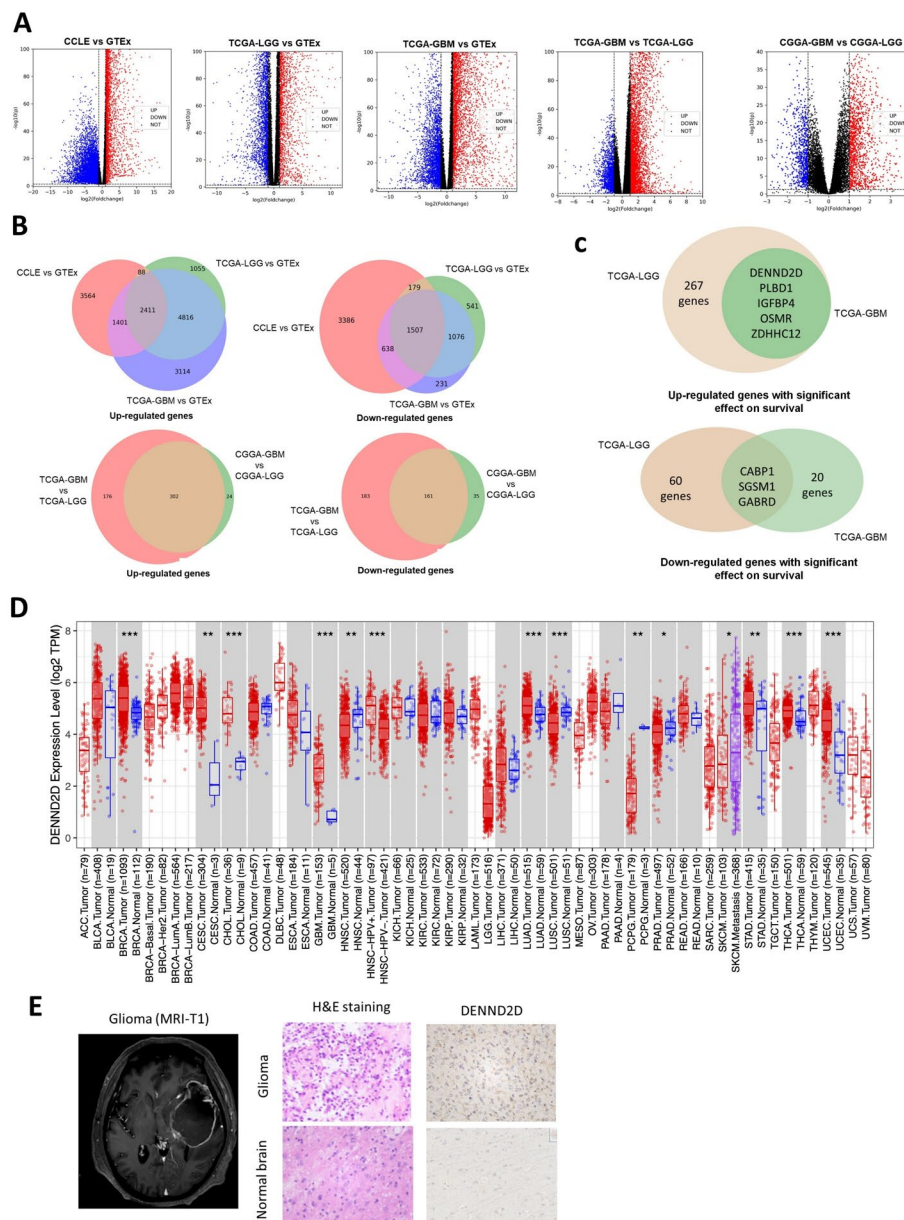


Fig. 1 Expression of *DENND2D* in glioma. **a** Screening of significantly differentially expressed genes across multiple databases. **b** Overlapping genes in the differentially expressed gene set. **c** Hub genes in overlapping genes. **d** Pan-cancer expression of *DENND2D*. **e** Expression of *DENND2D* in glioma and normal brain tissues

downregulated genes exhibiting significant differential expression across all analyses (Fig. 1b).

Log-rank tests and univariate Cox proportional hazards regression were applied to the 302 upregulated and 161 downregulated genes in the LGG and GBM cohorts. In LGG, 272 upregulated and 63 downregulated genes significantly correlated with survival outcomes. Strikingly, only 5 upregulated genes (*PLBD1*, *IGFBP4*, *OSMR*, *ZDHHC12*, and *DENND2D*) and 23 downregulated genes demonstrated prognostic significance in GBM. Notably, all five survival-associated genes upregulated in GBM were within the 272-significant LGG-upregulated set. Only three downregulated genes (*CABP1*, *SGSM1*, and *GABRD*) significantly correlated with survival in both LGG and GBM (Fig. 1c).

Following literature review and pan-cancer expression analysis of the five upregulated and three downregulated survival-associated DEGs, *DENND2D* was identified as the key candidate. *DENND2D* exhibited the highest HR (Supplementary Material S3). It exhibits tumor type-dependent dysregulation—significantly elevated in malignancies, including breast cancer, cervical carcinoma, cholangiocarcinoma, and GBM, yet decreased in head/neck squamous carcinomas and lung squamous cell carcinoma [10–14] (Fig. 1d). Immunohistochemical validation confirmed pronounced *DENND2D* overexpression in glioma specimens compared to normal brain tissue in GBM sample. This paradoxical expression pattern (contradicting its tumor-suppressive role in certain cancers) and the unresolved mechanistic contributions to glioma genesis prompted its selection as the primary focus for subsequent mechanistic investigations (Fig. 1e).

3.2 Expression of *DENND2D* in subgroups

Within TCGA dataset, *DENND2D* expression was significantly elevated in GBM compared to LGG. Further stratification revealed that, within LGG, tumors harboring an IDH mutation or 1p/19q co-deletion exhibited significantly reduced *DENND2D* expression relative to their wild-type counterparts. The status of MGMT promoter methylation also influenced *DENND2D* expression differentially: MGMT-methylated LGG samples showed significantly lower *DENND2D* levels than unmethylated LGG samples, whereas in GBM, no significant difference in *DENND2D* expression was observed based on MGMT methylation status. This distinctive expression pattern was validated in the CGGA-LGG cohort, where samples with IDH1 mutation or 1p/19q co-deletion showed significantly decreased *DENND2D* expression (Fig. 2a).

3.3 Correlation between *DENND2D* and clinical characteristics

Patients in the combined TCGA cohort were stratified into high- and low-expression groups based on the median *DENND2D* expression levels. High *DENND2D* expression was associated with significantly shorter OS compared to the low-expression group in both TCGA-GBM and TCGA-LGG cohorts. This negative prognostic association persisted in LGG across all molecularly stratified subgroups: patients with high *DENND2D* expression exhibited significantly poorer survival, irrespective of IDH mutation status or the absence of 1p/19q co-deletion. The only exception was observed in the 1p/19q co-deleted LGG subgroup, where survival did not differ significantly between high and low *DENND2D* expressers. Regarding MGMT status, high *DENND2D* expression correlated with significantly shorter survival in LGG patients, regardless of methylation status,

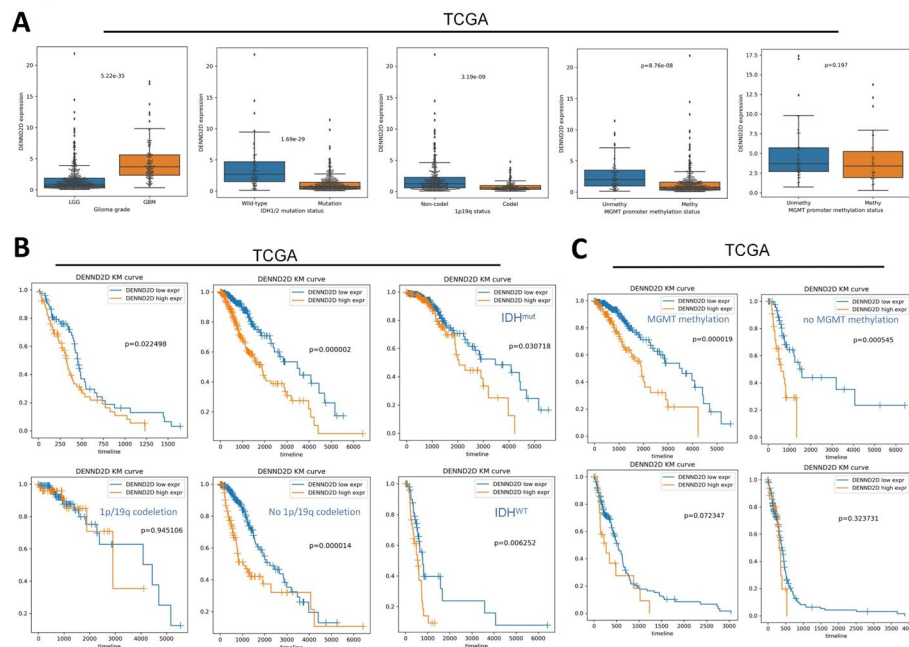


Fig. 2 Prognostic value of *DENND2D* in glioma. **a** Expression of *DENND2D* in different genotypes of glioma. **b** Survival differences among patients with different glioma genotypes with high and low expression of *DENND2D*

whereas in GBM, no significant survival difference was observed between *DENND2D* expression groups stratified by MGMT methylation (Fig. 2b, c).

3.4 Expression regulation of *DENND2D*

Methylation levels and CNV in genes have been demonstrated to influence their expression levels. In the context of gliomas, our research revealed a regulatory mechanism involving the expression of *DENND2D*, which is modulated by two specific methylation sites, cg00619207 and cg24641737. Our findings showed a negative correlation between expression and methylation levels, suggesting a potential relationship between DNA methylation and *DENND2D* expression in glioma tissues (Fig. 3a). However, this effect was more pronounced in LGG (stronger negative correlation and steeper fitting slope) than in GBM. The effects of CNV also vary: in LGG, samples with diploid or amplified copy numbers showed significantly higher expression levels than those with shallow deletions, and highly methylated samples were more frequently associated with shallow deletions, while low-methylated samples were more frequently associated with amplifications. In the context of GBM, diploid samples demonstrated significantly elevated expression levels compared with shallow deletion samples. However, no substantial disparities in expression were observed between diploid and amplification samples or between amplification and shallow deletion samples (Fig. 3b). Using the MEXPRESS tool, we found that in LGG, *DENND2D* expression exhibited a significant negative correlation with methylation, a significant positive correlation with copy number, and a significant negative correlation with OS. Conversely, in GBM, there was a significant negative correlation with methylation, with no substantial association observed with copy number or OS. A comprehensive analysis employing the DiseaseMeth database revealed that there was no statistically significant difference in *DENND2D* methylation

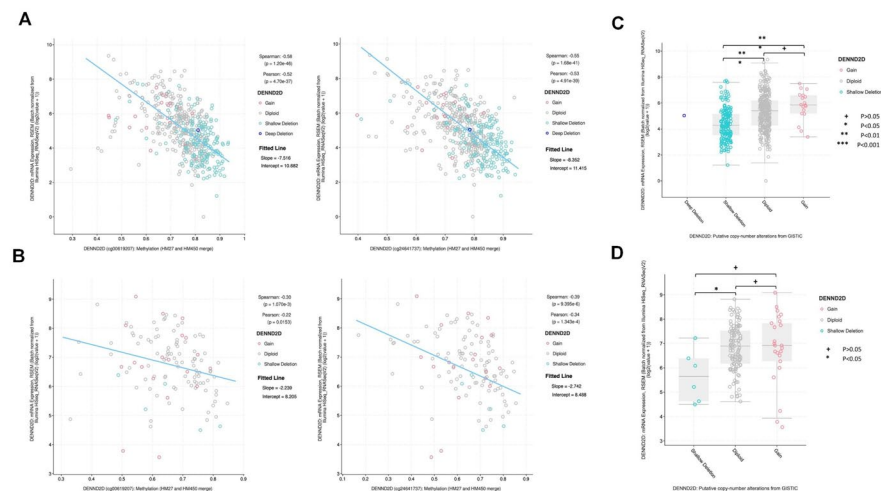


Fig. 3 Correlation between expression, methylation, and copy number of *DENND2D*. **a** Correlation between *DENND2D* expression and cg00619207/cg24641737 loci methylation in LGG. **b** Correlation between *DENND2D* expression and cg00619207/cg24641737 loci methylation in GBM. **c** Correlation between *DENND2D* expression and copy number variation in LGG. **d** Correlation between *DENND2D* expression and copy number variation in GBM

levels between LGG and GBM cases and normal brain tissue ($p < 0.05$ and methylation change > 0.2) (Fig. 3c, d).

3.5 Functional enrichment analysis of DENND2D

To investigate the role of *DENND2D* in glioma, we performed WGCNA analysis on TCGA-LGG and TCGA-GBM datasets. *DENND2D* was identified in module 8, which contained 1,263 genes in LGG, and in module 14, which contained 850 genes in GBM (Fig. 4a). Functional enrichment analysis (KEGG and GO) revealed that both modules were significantly enriched in pathways related to immune response and leukocyte activation (Fig. 4b). Subsequent analysis of the STRING protein interaction network identified key hub genes that directly interacted with *DENND2D* within each module. A comparison of the two datasets revealed the presence of six overlapping genes (*RAC2*, *TLR2*, *ITGB2*, *HCST*, *PTPRC*, and *HLA-DPA1*) in both LGG and GBM *DENND2D* hub genes. An in-depth pathway analysis of these six genes (via GeneCards) clearly demonstrated their common and significant involvement in immune system function, with core pathways including the Immune System, Adaptive Immune System, and Innate Immune System, and related core biological processes involving leukocyte adhesion, T cell proliferation regulation, and immune response (Fig. 4c, d). These results strongly suggest that *DENND2D* plays a role in immune regulation in gliomas.

In TCGA and CGGA datasets, we analyzed genes that exhibited significant positive and negative correlations with *DENND2D* expression. Functional enrichment analysis of the positively correlated genes (increased expression with *DENND2D* levels) in the LGG, GBM, and CGGA datasets consistently demonstrated that these genes were predominantly enriched in immune-related biological processes (GO) and pathogen infection-related pathways (KEGG), suggesting that high *DENND2D* expression is associated with enhanced immune activation (Fig. 5a). In contrast, functional enrichment analysis of negatively correlated genes (expression decreased with increasing *DENND2D*) revealed significant enrichment in synapse formation, transmission, and ion channel-related

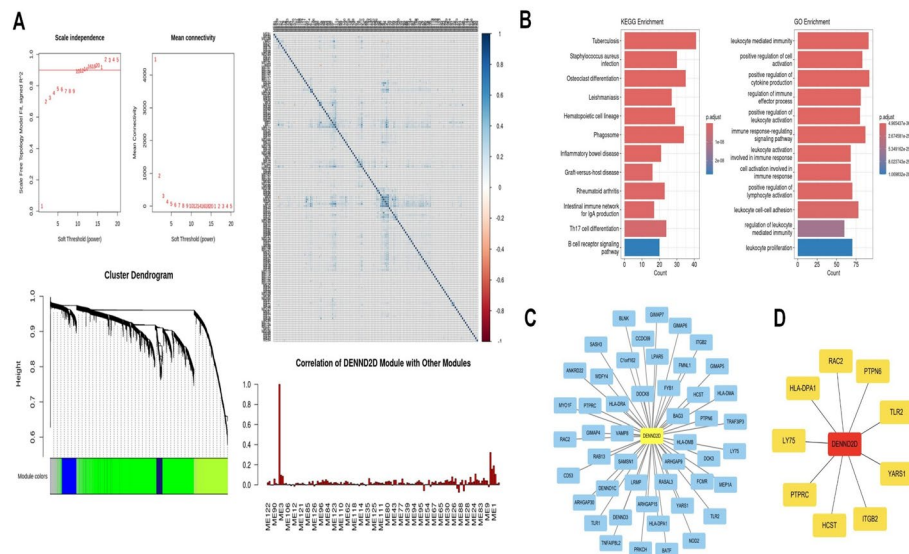


Fig. 4 WGCNA analysis of *DENND2D* in LGG. **a** WGCNA analysis of *DENND2D* in LGG. **b** KEGG and GO enrichment analysis of the module containing *DENND2D*. **c** PPI network of *DENND2D*-related proteins. **d** Top nine hub genes ranked by relevance

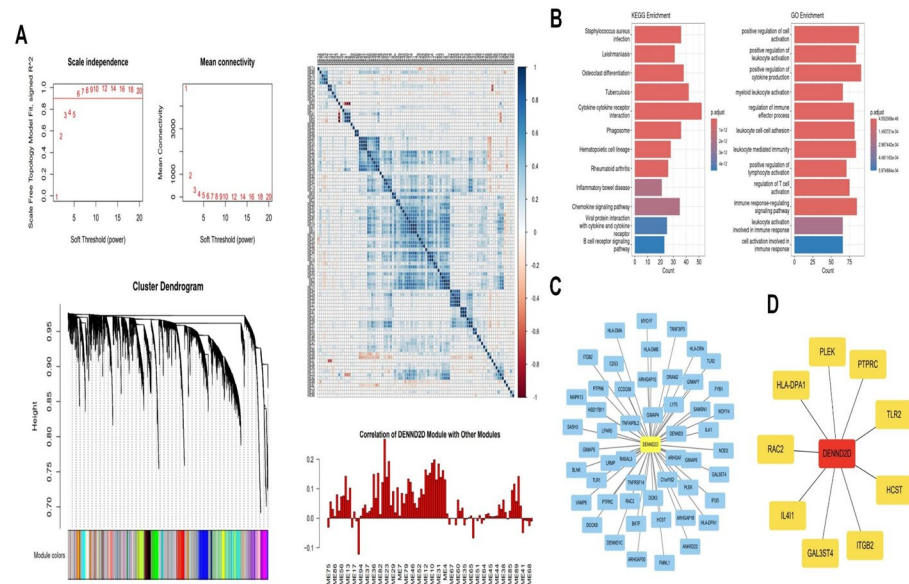


Fig. 5 WGCNA analysis of *DENND2D* in GBM. **a** WGCNA analysis of *DENND2D* in GBM. **b** KEGG and GO enrichment analysis of the module containing *DENND2D*. **c** PPI network of *DENND2D*-related proteins. **d** Top nine hub genes ranked by relevance

pathways (GO and KEGG) in LGG and CGGA data (Fig. 5b, c). These findings suggested that high *DENND2D* expression inhibits neuronal synaptic function. The findings from these two studies, which were independent and used different datasets, were highly consistent and support the important role of *DENND2D* in gliomas through positive regulation of immune responses and negative regulation of synaptic function.

3.6 The effect of DEENN2D on the immune microenvironment of glioma

A comprehensive investigation was conducted using CIBERSORT analysis of TCGA-LGG, TCGA-GBM, and CGGA data. This analysis focused on the infiltration of 22 types of immune cell types. The results revealed that *DENND2D* expression exhibits molecular subtype-specific associations with specific immune cell types. In TCGA-LGG and CGGA, *DENND2D* expression showed significant positive correlations with macrophages (M0, M1, M2), CD8 T cells, and $\gamma\delta$ T cells (Fig. 6a-d). However, *DENND2D* expression also showed significant negative correlations with memory B cells, resting NK cells, and naive CD4 T cells. In the TCGA-GBM cohort, *DENND2D* expression exhibited a significant positive correlation with M2 macrophages and T $\gamma\delta$ cells, a significant negative correlation with resting NK cells, and a significant negative correlation with M0 macrophages (contrary to the LGG cohort). Additionally, *DENND2D* expression

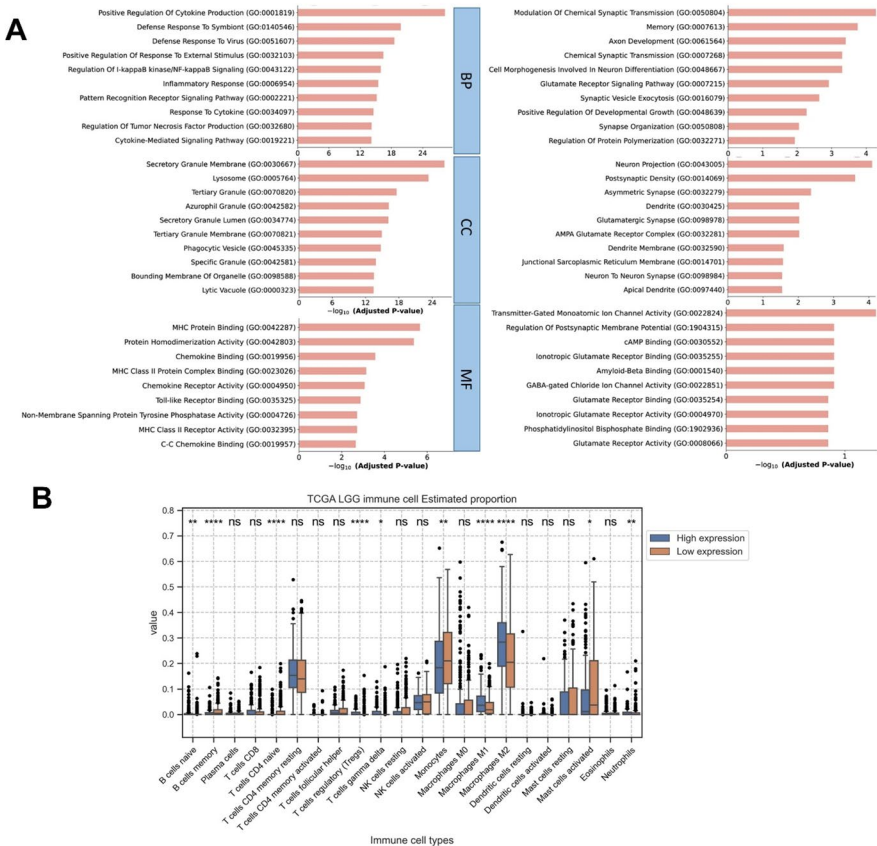


Fig. 6 Function analysis of *DENND2D*-related genes. **a** GO term enrichment of six overlapping hub genes in GeneCards. **b** Pathways enrichment of six overlapping hub genes in GeneCards. **c** GO analysis of positively and negatively correlated co-expressed genes of *DENND2D* in LGG. **d** Differential analysis of immune cell infiltration between high and low *DENND2D* expression groups in LGG

demonstrated a significant positive correlation with memory B cells and monocytes (in contrast to that in the LGG cohort) (Fig. 7a-c). Subsequent differential analysis further revealed that after grouping by the median expression of *DENND2D*, the infiltration proportions of M1 and M2 macrophages were significantly higher in the high-expression group of *DENND2D* in LGG and CGGA (Fig. 8a, b). In contrast, fewer immune cell types showed significant differences between high- and low-expression groups in GBM (Fig. 8c). These results suggested that *DENND2D* may influence the glioma microenvironment by regulating the infiltration of specific immune cells (especially macrophages), and that its mode of action differs significantly between LGG and GBM.

4 Discussion

DENND2D is a member of the DENN/MADD protein family and has recently emerged as a regulator of diverse tumorigenic processes. However, its potential role in glioma warrants further investigation. This study directly explored the function of *DENND2D* in glioma and its interplay with critical signaling pathways, providing crucial insights into its underlying mechanisms. Previous studies have demonstrated complex, context-dependent regulatory functions of *DENND2D* in various cancers. Downregulation of *DENND2D*, which is frequently mediated by promoter hypermethylation, correlates with poor prognosis in esophageal squamous cell carcinoma. Previous studies have evaluated the impact of *DENND2D* expression and methylation status on patient outcomes. In hepatocellular carcinoma (HCC), promoter hypermethylation-induced *DENND2D*

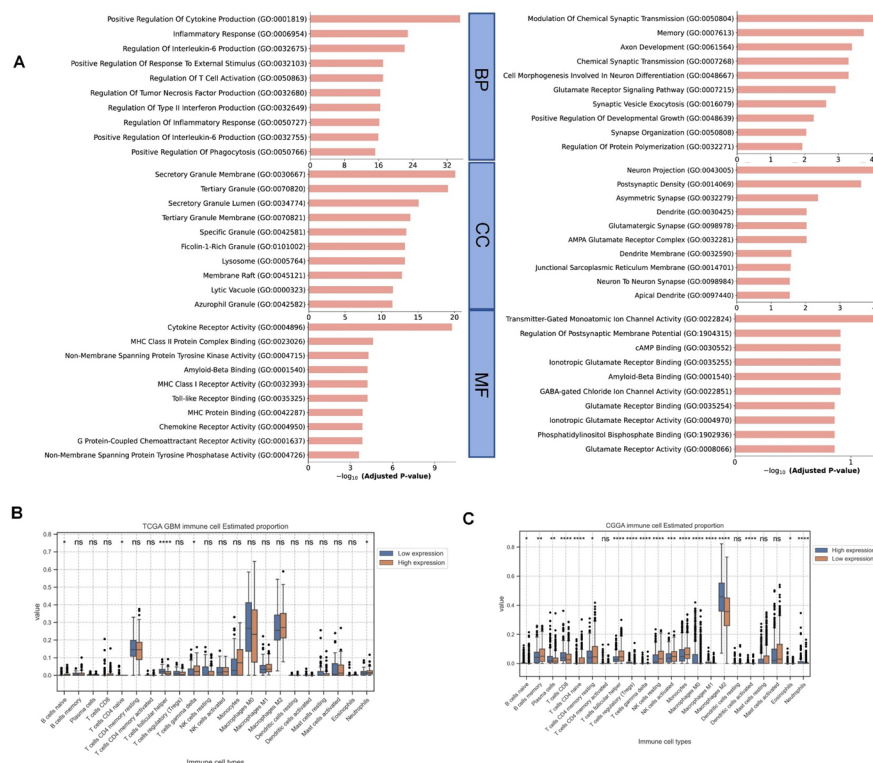


Fig. 7 Function analysis of *DENND2D* co-expressed genes. **a** GO analysis of positively and negatively correlated co-expressed genes of *DENND2D* in GBM. **b** Differential analysis of immune cell infiltration between high and low *DENND2D* expression groups in GBM. **c** Differential analysis of immune cell infiltration between high and low *DENND2D* expression groups in CGGA

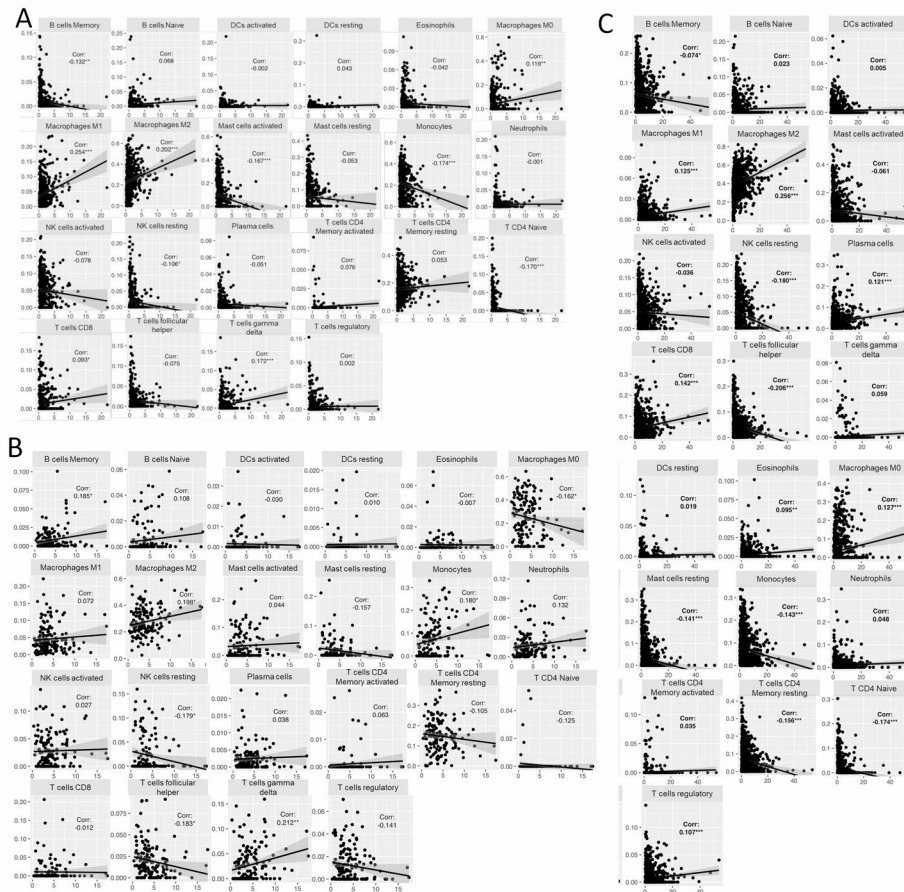


Fig. 8 Association between *DENND2D* expression and tumor microenvironment in glioma. **a** Correlation between *DENND2D* expression and immune cell infiltration in LGG. **b** Correlation between *DENND2D* expression and immune cell infiltration in GBM. **c** Correlation between *DENND2D* expression and immune cell infiltration in CGGA

downregulation is associated with early recurrence. In non-small cell lung cancer, miR-522-mediated downregulation targets *DENND2D* to suppress proliferation and metastasis. *DENND2D* itself is frequently downregulated, potentially acting as an early event that inhibits NSCLC proliferation and tumorigenicity [10, 13]. Conversely, exosomal miR-1246 promotes cell motility and invasion in oral squamous cell carcinoma by modulating *DENND2D* [14]. *DENND2D* has been identified as a prognostic biomarker for biochemical recurrence in prostate cancer [11]. In colon cancer, stage IV patients lacking *DENND2D* expression exhibited greater benefit from neoadjuvant chemotherapy, suggesting that its expression state influences the therapeutic response [15]. Collectively, these findings indicated that *DENND2D* can function as a tumor suppressor, and that its downregulation promotes oncogenesis or participates in oncogenic processes regulated by factors such as miRNAs.

Glioma, the most common primary malignant tumor of the CNS, is characterized by high recurrence rates and poor prognosis [16]. Its pathogenesis involves the dysregulation of key signaling pathways governing proliferation, migration, invasion, angiogenesis, and stemness [17–18]. As a membrane-trafficking protein, *DENND2D* may influence the localization or activity of the components within these pathways. For instance, the miR-1246/*DENND2D* axis, which promotes motility in OSCC, has potential roles in

cytoskeletal remodeling and migration pathways relevant to glioma. Glioma progression is also significantly influenced by the tumor microenvironment, including stiffened collagen networks facilitating proliferation and invasion and the pro-tumoral activity of tumor-associated macrophages (TAMs) [19]. TAMs secrete cytokines and chemokines that promote tumor growth, angiogenesis, metastasis, and immunosuppression, whereas neutrophils exhibit context-dependent roles modulated by hypoxia. Aberrant *DENND2D* expression or function could thus impact tumor cell interactions with the TME (tumor microenvironment), affecting adhesion, motility, or response to microenvironmental cues. Our GO and KEGG pathway analyses of *DENND2D*-correlated genes revealed significant enrichment of upregulated genes in immune/inflammatory pathways. Downregulated genes were associated with synaptic signaling and neuronal communication, suggesting *DENND2D* may suppress normal neuronal function while promoting an immune-reactive TME. This duality aligns with emerging evidence that gliomas remodel neuronal circuits to support growth while evading immune surveillance. Notably, the overlap of *DENND2D*-correlated genes with pathogen response pathways is intriguing, given the recent links between endogenous retroviruses, viral mimicry, and glioma immunogenicity. *DENND2D* participation in these processes could represent a novel mechanism through which gliomas co-opt immune pathways for survival.

A key finding of our study was the strong association between *DENND2D* expression and immune cell infiltration in gliomas. WGCNA placed *DENND2D* within a module highly enriched for immune-related pathways, including leukocyte activation, inflammatory responses, and defense responses to viruses. Genes exhibiting the highest co-expression with *DENND2D* were implicated in immune regulation, indicating a potential role for *DENND2D* as an immune modulator in glioma. CIBERSORT-based immune cell deconvolution further supported this hypothesis, revealing distinct immune landscapes in *DENND2D*-high gliomas. In LGG, *DENND2D* expression positively correlated with M1 and M2 macrophage infiltration, CD8 + T cells, and $\gamma\delta$ T cells, while showing negative correlations with B cells and resting NK cells. In GBM, *DENND2D* exhibited a strong positive correlation with M2 macrophages but a negative correlation with M0 macrophages (undifferentiated precursors). This pattern suggests *DENND2D* may contribute to an immunosuppressive TME by promoting polarization towards the pro-tumoral M2 macrophage phenotype, a hallmark of immune evasion in gliomas [20]. Differences in immune correlations between LGG and GBM likely reflect their inherent immunological heterogeneity. Given the increasing focus on glioma immunotherapy, our findings position *DENND2D* as a potential mediator of resistance to immune checkpoint blockades. Future studies should explore whether targeting *DENND2D* can enhance T-cell activation or reprogram macrophages towards an antitumor phenotype.

There are some limitations of this study. First, the expression pattern and prognostic value of *DENND2D* were primarily analyzed using public bioinformatics databases. We plan to validate *DENND2D* expression levels and their correlation with clinicopathological features in our own cohort of glioma clinical tissue samples in subsequent studies. Second, although our analyses strongly suggest an oncogenic role for *DENND2D* in glioma, its specific functional mechanisms, particularly its contribution to immune modulation, remain unclear. We plan to perform cellular and animal experiments, including *DENND2D* knockdown and overexpression, to clarify its functional impact on glioma

progression and immune regulation. Third, although we observed significant correlations between *DENND2D* expression and specific immune cell infiltrates, the resolution of immune cell deconvolution was limited, and the precise nature of these immune interactions requires further investigation. Further studies using techniques such as single-cell sequencing or multiplex immunofluorescence are needed to confirm these associations and understand the functional state of these immune cells. Finally, although *DENND2D* has been proposed as a potential therapeutic target, its modulatory effects have not yet been explored. In future studies, we will investigate the feasibility of targeting *DENND2D* and assess its impact, particularly on overcoming immunotherapy resistance, using preclinical models.

Gene dysregulation in gliomas extends to non-coding RNAs (ncRNAs), including lncRNAs and miRNAs [21, 22], which modulate target gene expression and influence tumor behavior. In LGG and GBM, *DENND2D* expression showed a negative correlation with methylation levels at two key CpG sites (cg00619207 and cg24641737). This suggested that hypomethylation may drive *DENND2D* upregulation in glioma. Additionally, post-translational modifications (PTMs) and liquid-liquid phase separation (LLPS) are recognized as key regulatory mechanisms in oncogenesis and therapy [22, 23]. PTMs influence protein activity, localization, and interactions, whereas LLPS governs cellular processes by compartmentalizing biomolecules within condensates. The *DENND2D*'s function and localization might be modulated by PTMs or LLPS, indirectly shaping its glioma-specific role. Future studies employing single-cell epigenomics or CRISPR-based methylation editing may elucidate the precise regulatory dynamics of *DENND2D* across glioma subtypes. By linking genomic alterations to immune escape mechanisms, *DENND2D* may offer a novel potential target to address the urgent need for innovative strategies in glioma therapy.

5 Conclusions

In summary, our study revealed the pro-tumor role of *DENND2D* in glioma. Its association with aggressive subtypes, immune infiltration, and poor prognosis suggested a potential use as a biomarker for patient stratification. Furthermore, the correlation between *DENND2D* expression and immune suppression suggested that targeting *DENND2D* may enhance the efficacy of immunotherapy. This correlation provides a foundation for future research.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1007/s12672-026-04743-5>.

Supplementary Material 1.
Supplementary Material 2.
Supplementary Material 3.

Author contributions

H.Y and L.Z designed the research study and analyzed the data. L.Z, Z.P and Y.W wrote the manuscript and revision. All authors contributed to editorial changes in the manuscript. All authors read and approved the final manuscript.

Funding

This study was supported by the Zhejiang Provincial Natural Science Foundation of China (No. LY18H160020).

Data availability

Data supporting the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

The study was conducted in accordance with the Declaration of Helsinki, and approved by the Medical Ethics Committee of The First Affiliated Hospital, Zhejiang University School of Medicine (#2019-52). The patient provided written informed consent for publication of this case report and the associated images.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Received: 13 July 2025 / Accepted: 2 March 2026

Published online: 08 March 2026

References

1. McKinnon C, Nandhabalan M, Murray SA, Plaha P. Glioblastoma: clinical presentation, diagnosis, and management. *BMJ*. 2021;374:n1560.
2. Omuro A, DeAngelis LM. Glioblastoma and other malignant gliomas: a clinical review. *JAMA*. 2013;310(17):1842–50.
3. Alexander BM, Cloughesy TF. Adult Glioblastoma. *J Clin Oncol*. 2017;35(21):2402–9.
4. Xu Q, Huang K, Meng X, Weng Y, Zhang L, Bu L, Zheng X, Cai J, Zhan R, Chen Q. Safety and Efficacy of Anlotinib Hydrochloride Plus Temozolomide in Patients with Recurrent Glioblastoma. *Clin Cancer Res*. 2023;29(19):3859–66.
5. Muhammad S, Maridevaru MC, Roy S, Chen D, Zeng W, Sun L, Zhang Y, Guo B. Piezodynamic Therapy: Unleashing Mechanical Energy and Featuristic Next Generation Therapeutic Paradigms for Glioblastoma. *ACS Nano*. 2025;19(37):33008–58.
6. Zhang L, Wang Y, Peng Z, Weng Y, Fang Z, Xiao F, Zhang C, Fan Z, Huang K, Zhu Y, Jiang W, Shen J, Zhan R. The progress of multimodal imaging combination and subregion based radiomics research of cancers. *Int J Biol Sci*. 2022;18(8):3458–69.
7. Cerami E, Gao J, Dogrusoz U, Gross BE, Sumer SO, Aksoy BA, Jacobsen A, Byrne CJ, Heuer ML, Larsson E, Antipin Y, Reva B, Goldberg AP, Sander C, Schultz N. The cBio cancer genomics portal: an open platform for exploring multidimensional cancer genomics data. *Cancer Discov*. 2012;2(5):401–4.
8. Xiong Y, Wei Y, Gu Y, Zhang S, Lyu J, Zhang B, Chen C, Zhu J, Wang Y, Liu H, Zhang Y. DiseaseMeth version 2.0: a major expansion and update of the human disease methylation database. *Nucleic Acids Res*. 2017;45(D1):D888–95.
9. Wang Y, Gao B, Zhang L, Wang X, Zhu X, Yang H, Zhang F, Zhu X, Zhou B, Yao S, Nagayama A, Lee S, Ouyang J, Koh SB, Eisenhauer EL, Zarrella D, Lu K, Rueda BR, Zou L, Su XA, Yeku O, Ellisen LW, Wang XS, Lan L. Meiotic protein SYCP2 confers resistance to DNA-damaging agents through R-loop-mediated DNA repair. *Nat Commun*. 2024;15(1):1568.
10. Kanda M, Nomoto S, Oya H, Takami H, Hibino S, Hishida M, et al. Downregulation of DENND2D by promoter hypermethylation is associated with early recurrence of hepatocellular carcinoma. *Int J Oncol*. 2014;44(1):44–52.
11. Chang CF, Chen LC, Chen YT, Huang CY, Yu CC, Lin VC, et al. Unveiling DENND2D as a Novel Prognostic Biomarker for Prostate Cancer Recurrence: From Gene to Prognosis. *Biomedicines*. 2024;13(1):25.
12. Hibino S, Kanda M, Oya H, Takami H, Shimizu D, Nomoto S, et al. Reduced expression of DENND2D through promoter hypermethylation is an adverse prognostic factor in squamous cell carcinoma of the esophagus. *Oncol Rep*. 2014;31(2):693–700.
13. Ling B, Zheng H, Fu G, Yuan J, Shi T, Chen S, et al. Suppression of non-small cell lung cancer proliferation and tumorigenicity by DENND2D. *Lung Cancer*. 2013;79(2):104–10.
14. Sakha S, Muramatsu T, Ueda K, Inazawa J. Exosomal microRNA miR-1246 induces cell motility and invasion through the regulation of DENND2D in oral squamous cell carcinoma. *Sci Rep*. 2016;6:38750.
15. Ma WJ, Chen Y, Peng JH, Tang C, Zhang L, Liu M, Hu S, Xu H, Tan H, Gu Y, Pan ZZ, Chen G, Zhou ZG, Zhang RX. Stage IV colon cancer patients without DENND2D expression benefit more from neoadjuvant chemotherapy. *Cell Death Dis*. 2022;13(5):439.
16. Xu Q, Huang K, Meng X, Weng Y, Zhang L, Bu L, et al. Safety and Efficacy of Anlotinib Hydrochloride Plus Temozolomide in Patients with Recurrent Glioblastoma. *Clin Cancer Res*. 2023;29(19):3859–66.
17. Song KW, Lim M, Monje M. Complex neural-immune interactions shape glioma immunotherapy. *Immunity*. 2025;58(5):1140–60.
18. Xiao F, Shen J, Zhou L, Fang Z, Weng Y, Zhang C, et al. ZNF395 facilitates macrophage polarization and impacts the prognosis of glioma. *Am J Cancer Res*. 2022;12(9):4312–25.
19. Xiang X, Wang J, Lu D, Xu X. Targeting tumor-associated macrophages to synergize tumor immunotherapy. *Signal Transduct Target Ther*. 2021;6(1):75.
20. Wu R, Sun C, Chen X, Yang R, Luan Y, Zhao X, Yu P, Luo R, Hou Y, Tian R, Bian S, Li Y, Dong Y, Liu Q, Dai W, Fan Z, Yan R, Pan B, Feng S, Wu J, Chen F, Yang C, Wang H, Dai H, Shu M. NSUN5/TET2-directed chromatin-associated RNA modification of 5-methylcytosine to 5-hydroxymethylcytosine governs glioma immune evasion. *Proc Natl Acad Sci U S A*. 2024;121(14):e232161121.
14. Wang Y, Mo Y, Gong Z, Yang X, Yang M, Zhang S, et al. Circular RNAs in human cancer. *Mol Cancer*. 2017;16(1):25.
21. Wang D, Zuo S, Ge J, Qu H, Wu J, Yi N, et al. circTP63-N suppresses the proliferation and metastasis of nasopharyngeal carcinoma via engaging with HSP90AB1 to modulate the YAP1/Hippo signaling pathway. *Sci China Life Sci*. 2025;68(3):689–705.
22. Yan X, Zhang M, Wang D. Interplay between posttranslational modifications and liquid–liquid phase separation in tumors. *Cancer Lett*. 2024;584:216614.

23. Chen H, Huang S, Quan L, Yu C, Zhu Y, Sun X, et al. Liquid–liquid phase separation: a potentially fundamental mechanism of sepsis. *Cell Death Discov.* 2025;11(1):310.

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.